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Numerical Methods for Elliptic Problems – Exercise Sheet 1

Due date: October 13, 2026, 10:15-11:45, Room SP2-0346

Exercise 1 (Vector identities, integration-by-parts):
Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with piecewise smooth boundary, $\vec{b} \in C^1(\bar{\Omega}; \mathbb{R}^d)$ and $u, v \in C^2(\bar{\Omega})$. Let $\vec{n}$ denote the outward unit normal and write $\partial_n u := \nabla u \cdot \vec{n}$. Show the following identities:

(a) $\operatorname{div}(\vec{b}\, u) = \vec{b} \cdot \nabla u + u \operatorname{div}(\vec{b})$
(b) $\int_\Omega \vec{b} \cdot \nabla u \, dx = -\int_\Omega u \operatorname{div}\vec{b} \, dx + \int_{\partial\Omega} u\, \vec{b} \cdot \vec{n} \, ds$
(c) $\int_\Omega \nabla u \cdot \nabla v \, dx = -\int_\Omega \Delta u \, v \, dx + \int_{\partial\Omega} \partial_n u \, v \, ds$

Hint: Use the product rule of differentiation and the Gauß integral theorem.

Exercise 2 (Fundamental lemma 1):
Prove the following assertion: Let $\Omega \subset \mathbb{R}^d$ be an open set and $f \in C(\overline{\Omega})$. Then

$$\int_\omega f(x)\, dx = 0 \quad \text{for every bounded measurable set } \omega \subset \Omega \quad \Longleftrightarrow \quad f \equiv 0.$$

Show also that if the assumption $f \in C(\overline{\Omega})$ is replaced by $f \in L^1(\Omega)$, then the equivalence remains valid with $f = 0$ almost everywhere in Ω.

Hint: It suffices to test closed balls contained in Ω.

Exercise 3 (Fundamental lemma, 2nd version):
Let $\Omega \subset \mathbb{R}^d$ be a bounded domain and $f \in C(\overline{\Omega})$. Show that

$$\int_\Omega f(x) v(x)\, dx = 0 \quad \text{for every } v \in C(\overline{\Omega}) \text{ with } v|_{\partial\Omega} = 0 \qquad \Longleftrightarrow \qquad f(x) = 0 \quad \text{for every } x \in \Omega.$$

Show moreover that for $f \in L^2(\Omega)$

$$\int_\Omega f v \, dx = 0 \quad \text{for all } v \in C_c^\infty(\Omega) \quad \Leftrightarrow \quad \int_\Omega f v \, dx = 0 \quad \text{for all } v \in L^2(\Omega) \quad \Leftrightarrow \quad f = 0 \quad \text{a.e. in } \Omega.$$

Hint: In the continuous case, assume $f(x_0) > 0$ at some $x_0 \in \Omega$ and construct a test function supported in a small ball around x_0. In the L^2 case, use the density of $C_c^\infty(\Omega)$ in $L^2(\Omega)$.

Exercise 4:
Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with Lipschitz boundary, $a \in L^\infty(\Omega)$ satisfy $a(x) \geq a_0 > 0$ almost everywhere for some constant a_0, and let $f \in L^2(\Omega)$.

(a) Derive the necessary optimality condition (Euler–Lagrange equation) for

$$J(u) := \frac{1}{2}\int_\Omega a|\nabla u|^2\,dx - \int_\Omega fu\,dx \longrightarrow \min_{u:u|_{\partial\Omega}=0}$$

in the form of a variational problem. Compare with the minimal surface problem.

(b) Use integration by parts and the fundamental lemma to derive the corresponding PDE and state its boundary condition assuming that $a \in C^1(\overline{\Omega})$, $f \in C(\overline{\Omega})$, and $u \in C^2(\overline{\Omega})$.

Exercise 5:
Let $\Omega \subset \mathbb{R}^d$ be a domain and $g = u_g|_{\partial\Omega}$ for some $u_g \in C^2(\overline{\Omega})$.

Show that any classical solution $u \in C^2(\Omega) \cap C(\overline{\Omega})$ of

$$-\Delta u = 0 \quad \text{in } \Omega, \qquad u = g \quad \text{on } \partial\Omega$$

can be decomposed as $u = u_g + w$, where w solves

$$-\Delta w = f \quad \text{in } \Omega, \qquad w = 0 \quad \text{on } \partial\Omega.$$

Determine the right-hand side f, which depends only on u_g.

Exercise 6:
Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with piecewise smooth boundary. Recall the diffusion–convection–reaction equation

$$-\operatorname{div}(a\nabla u - \vec{b}\,u) + cu = f \quad \text{in } \Omega.$$

Assume sufficient smoothness for the following classical calculations, in particular $\vec{b} \in C^1(\overline{\Omega}; \mathbb{R}^d)$. Show that for $a \equiv 0$ this can be rewritten as a transport equation

$$\vec{b}' \cdot \nabla u + c'u = f' \quad \text{in } \Omega.$$

Assume $f' \equiv c' \equiv 0$ and $\vec{b}'$ constant. Describe the solutions along connected characteristic segments in Ω.

Extra: Use this description and a sketch to formulate boundary conditions that potentially lead to a well-posed problem.

Exercise 7:
Try to install NGSolve on your system by using pip or by compiling it from source such that the `CppExtensionTest.ipynb` notebook can be run successfully. Installation information can be found in `getting-started/ngsolve-installation.pdf`.

Ex 1

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Exercise 1 (Vector identities, integration-by-parts):
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Let $\vec{n}$ denote the outward unit normal and write $\partial_n u := \nabla u \cdot \vec{n}$. Show the following identities:

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Hint: Use the product rule of differentiation and the Gauß integral theorem.

Exercise 2 (Fundamental lemma 1):
Prove the following assertion: Let $\Omega \subset \mathbb{R}^d$ be an open set and $f \in C(\overline{\Omega})$. Then

$$\int_\omega f(x)\,dx = 0 \quad \text{for every bounded measurable set } \omega \subset \Omega \quad \overset{①}{\iff} \quad f \equiv 0.$$

② Show also that if the assumption $f \in C(\overline{\Omega})$ is replaced by $f \in L^1(\Omega)$, then the equivalence remains valid with $f = 0$ almost everywhere in Ω.

Hint: It suffices to test closed balls contained in Ω.

① $\Leftarrow$ clear

$\Rightarrow$

Assume for the sake of contradiction there ex $x_0 \in \Omega$ with $f(x_0) \neq 0$. Without loss $f(x_0) > 0$ (Ω is open!)
Due to the continuity of f, there ex. $\delta > 0$ s.t. $f(x) > 0$ for $x \in B_\delta(x_0)$ with δ sufficiently small
Now choose $r < \delta$ s.t. $\overline{B_r(x_0)} \subset \Omega$. It holds:

$$0 = \int_{B_r(x_0)} f(x)\,dx \geq \int_{B_\delta(x_0)} \underbrace{f(x)}_{>0}\,dx > 0 \quad \lightning$$

↑ special case of ω

② $\Leftarrow$ $f \equiv 0$ a.e $\Rightarrow \int_\omega f\,dx = 0$ (clear)

$\Rightarrow$ We show that $\int_N f(x)\,dx = 0$ for all null sets N.
For this choose a subset on which f is strictly positive:

$$E_{n,R} = \{x \in \Omega \cap B_R(0) : f(x) \geq \tfrac{1}{n}\}.$$

Clearly for $x \in E_{n,R}$: $f(x) \geq \frac{1}{n} > 0 \quad \forall n, R$ by definition.
Also $E_{n,R}$ measurable (clear) and bounded due to "$\cap B_R(0)$" for $R < \infty$. Now,

$$0 \overset{Ass}{=} \int_{E_{n,R}} f(x)\,dx \Rightarrow \mu(E_{n,R}) = 0$$

Analogously, for $E'_{n,R} = \{x \in \Omega \cap B_R(0) : f(x) \leq -\tfrac{1}{n}\}$ it follows that $f(x) \leq -\frac{1}{n} < 0$, $\mu(E'_{n,R}) = 0$.
Now,

$$A_{n,R} = E_{n,R} \cup E'_{n,R}$$

and

$$B = \bigcup_{n,R} A_{n,R}, \quad \mu(B) \overset{?}{=} 0$$

Further,

$$\int_N f(x)\,dx \leq \int_B f(x)\,dx = 0 \quad \forall \text{ null sets } N$$

Why? For every x st $f(x) > 0$ (or < 0), there ex a n, R s.t. $x \in A_{n,R}$

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